AAE / ME 538: Air-Breathing Propulsion

Syllabus

Prof. Guillermo Paniagua, Carson D. Slabaugh,

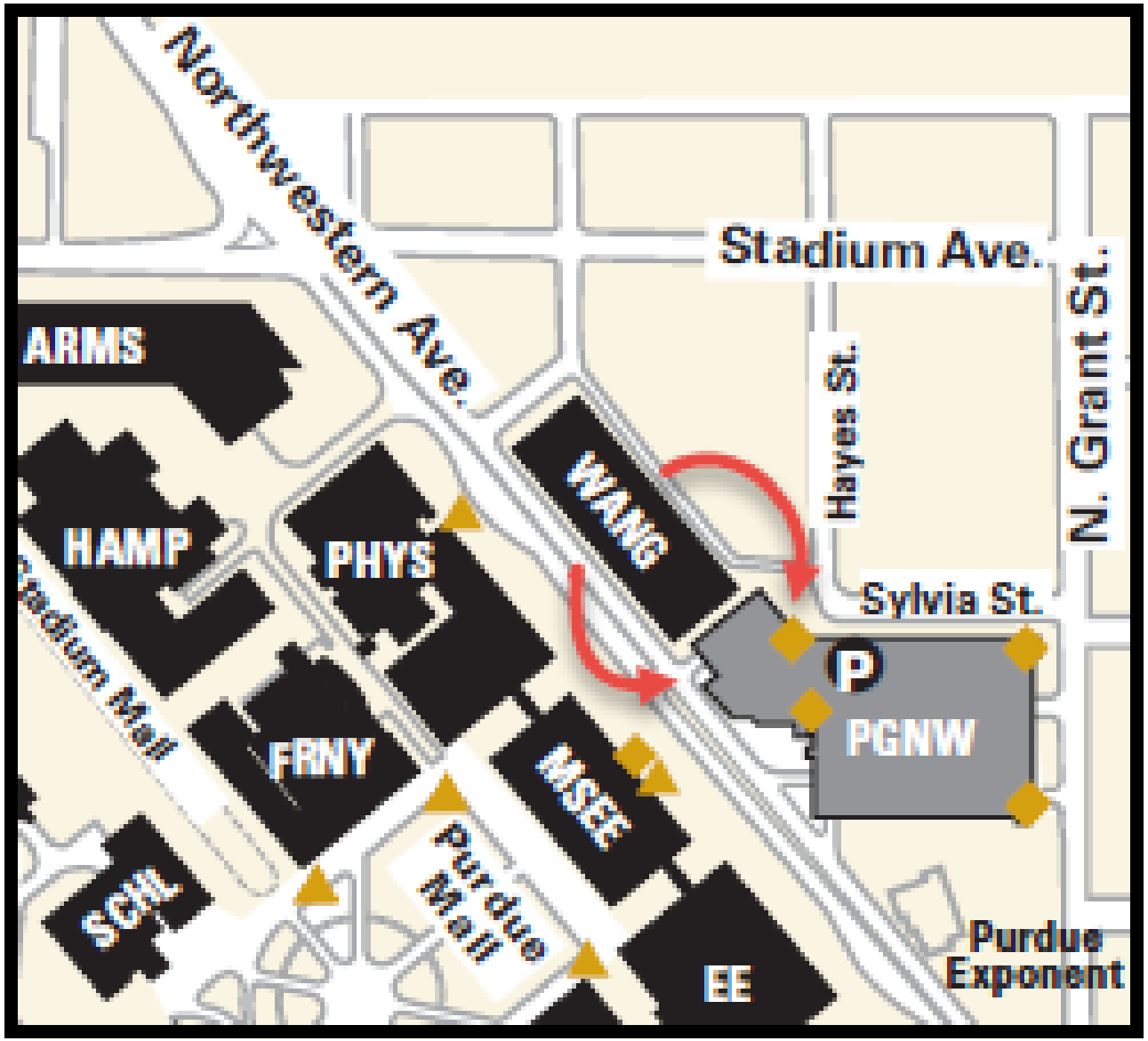
School of Aeronautics and Astronautics
School of Mechanical Engineering
Maurice J. Zucrow Laboratories
Purdue University



Emergency plan

- Tornado where to go
- Fire – no elevator
- Shelter In-Place – how to lock

- https://mediaspace.itap.purdue.edu/media/Orientation+Video+Part+4_Physical+Classroom+information+regarding+Emergencies/1_8cjqsi76



Audience: Engineers concerned with efficient aviation air transportation, and power generation. Researchers aiming to be trained in the development of aero-engines. Engineering students interested in pursuing a career in the aircraft propulsion and advanced energy systems.

Class Time & Location:

Monday	4:30pm-5:45pm	Wang 2579
Wednesday	4:30pm-5:45pm	Wang 2579

Required Textbook: no specific textbook for this course

Good References

Kurzke, Halliwell, Hill – “Propulsion and Power”
von Karman Lecture Series – “Aeroengine design”
Saravanamuttoo - "Gas Turbine Theory”
Internal Flow – Greitzer
Mattingly – “Elements of Propulsion”
Heiser and Pratt

Office hours: Six office hours / week

Instructor: Guillermo Paniagua, MS Teams – “gpaniag“
Office hours: Tuesday 9:00am - 10:30am MS Teams online & Zucrow ZL9

Lead teaching assistant: Andrea Ruan, MS Teams/email “ruana@purdue.edu”
Office Hours: Monday 6pm – 7:30pm MS Teams online & In-person Campus

Secondary Teaching assistants:
1. Andrej Damjanov, MS Teams/email – “adamjan@purdue.edu”
Office Hours: Tuesday 11am – 12:30pm MS Teams online & In-person Campus

2. Khoi Leong, MS Teams/email - “leong17@purdue.edu“
Office hours: Thursday 6pm – 7:30pm MS Teams online & In-person Campus

We will be online every week, waiting for your questions.
To ensure a prompt response please contact us through MS Teams.
If you have an Urgent request, please call us using MSTeams

Objectives and prerequisites

By the end of the course you should be able to:

1. Design the airbreathing propulsion system for any vehicle. Provide the overall engine architecture, size, engine matching operating conditions. *Methods of Evaluation: Team engine design project*
2. Define the thermodynamic cycle and size for an aircraft mission. *Methods of Evaluation: Quiz and Team engine design project*
3. Analyze the different engine components, and engine-aircraft interaction on intake. *Methods of Evaluation: Quiz and Team engine design project*
4. Communicate final developed design to a panel of experts in the field . *Methods of Evaluation: Team engine design project, Peer evaluations, group discussions*
5. Develop a critical analysis to optimize future engine concepts based on physical principles. *Methods of Evaluation: Concept proposal, quizzes*
6. Outline future research programs. *Methods of Evaluation: Quiz.*

Prerequisites:
- Fundamentals of thermodynamics, such as ME200
- Fundamentals of fluid mechanics, such as ME309

Schedule

Dates	#	Section/subsection
8/25/25	1	**Introduction**
8/27/25		
9/03/25		**1.Compressible flow fundamentals**
9/08/25	3	
		2.Subsonic propulsion systems:
9/10/25		
9/15/25		2.a gas engine cycle and performance criteria
9/17/25	3	
9/22/25		
9/24/25		2.b compression systems intake, fan and compressor
9/29/25	3	
10/1/25		
10/6/25		2.c combustors
10/8/25	3	
10/15/25		
10/20/25		2.d expansion systems: turbine and nozzles
10/22/25		
10/27/25	3	
10/29/25	*1*	*First presentation of the project*
11/3/25	1	2.e design point analysis of real engines
11/5/25	1	2.f off-design analysis
		3.Supersonic propulsion systems
11/10/25		3.a Introduction to High Speed ABP
11/12/25	2	
11/17/25	1	3.b compression system
11/19/25	1	3.c combustors
11/24/25	*1*	*Final presentation of the project*
12/1/25	1	3.d expansion system
12/3/25	1	3.e design point analysis of ideal and real engines
		4. Future perspectives and innovative architectures
12/8/25	1	4 New architectures for higher efficiency & Pressure Gain Combustion

Important days:

Labor day Sept 1

Fall break Oct 13-14

Thanksgiving Nov 26-29

Dead week: Dec 8-12

Finals week: Dec 15-19

Quizzes:

Quiz 1 - 10/06/2025

Quiz 2 - 11/05/2023

Quiz 3 - 12/03/2023

Project:

1st presentation: 10/29/25

Final presentation: 11/24/25

Final report: 12/15/25

Notes, Homework and Quizzes

Estimated Effort: 6hours per week.

Lecture Notes: Notes will be shared in BrightSpace. These notes provide a foundation for the topics covered in lecture.

Homework Policy:

Homework assignments are always due at the beginning of class, two class periods after they are assigned (not the immediate next class).

For example: homework assigned on a Monday is due at the beginning of class on the following Monday.

No exceptions: homework must be submitted at the start of class; late submissions will not be accepted.

Quiz: There are 3 quizzes throughout the semester. The quizzes are closed book and closed notes. If needed formula sheets will be provided.

Engine design project

Problem statement, and boundary conditions are provided in the document:

"*ProjectDesign2015_RFP.pdf*" – Form a team ASAP, I will release this week

Deliverables include:

- First and Second presentations
- Final report (25 pages max) describe your methodology, solution

Team project on the design of a new engine. Each team should have 5 members, 6 are accepted if they include online students, Deadline to have team 8/29/2025

You should use GasTurb 15 to analyze numerically the gas turbine engine. GasTurb has been purchased, and instructions for installation will be provided by the lead TA

Grades: Rubric for project presentation

PI – Performance Indicator	Proficient 3	Emerging 2	Developing 1	Deficient 0
1. **Context of and Purpose** Includes considerations of audience, purpose, and the circumstances surrounding the writing task(s)	Demonstrates a thorough understanding of context, audience, and purpose that is responsive to the assigned task(s) and focuses all elements of the work	Demonstrates adequate consideration of context, audience, and purpose and a clear focus on the assigned task(s) (e.g., the task aligns with audience, purpose, and context)	Demonstrates awareness of context, audience, purpose, and to the assigned task(s) (e.g., begins to show awareness of audience's perceptions and assumptions)	Does not demonstrate awareness of context, audience, purpose, and to the assigned task(s)
2. **Content Development** Consistently appropriate; analysis is logical and sound – no gaps in topic coverage; data / analysis clearly support the thesis	Uses appropriate, relevant, and compelling content to illustrate mastery of the subject, conveying the writer's thorough understanding and analysis	Uses appropriate, relevant, and compelling content to explore ideas and displays an adequate understanding and analysis of the topic(s)	Uses appropriate and relevant content to develop and explore ideas of the topic(s) with minimal understanding and analysis	Does not use appropriate and relevant content to develop and explore ideas and does not demonstrate understanding or analysis of the topic(s)
3. **Convention & Mechanics** Follows any formal and informal rules inherent in the expectations for technical writing for the discipline; proper grammar, language, spelling, content flow, formatting, and follows assignment requirements	Demonstrates detailed attention to and successful execution of a wide range of conventions particular to a specific discipline and/or writing task(s) including organization, content, presentation, formatting, and stylistic choices; uses language that skillfully conveys meaning with clarity and fluency, and is virtually error-free	Demonstrates consistent use of important conventions particular to a specific discipline and/or writing task(s), including organization, content, presentation, and stylistic choices; uses language that generally conveys meaning; has few errors	Follows expectations appropriate to a specific discipline and/or writing task(s) for basic organization, content, and presentation: uses language that minimally conveys meaning to readers; may include some errors	Does not follow expectations appropriate to a specific discipline and/or writing task(s) for basic organization, content, and presentation; does not use language that conveys meaning to readers; includes numerous errors
4. **Sources and Evidence** Identify, retrieve, organize, and effectively use the needed information to support the central message; credit is given for all work, using standard format	Demonstrates skillful use and citing of high-quality, credible, relevant sources to develop ideas that are appropriate for the discipline and genre	Demonstrates consistent citation of and use of credible, relevant sources to support ideas that are situated within the discipline and genre citing all sources	Demonstrates an attempt to use and cite credible and/or relevant sources to support ideas that are appropriate for the discipline and genre	Does not demonstrate an attempt to use or cite credible and/or relevant sources to support ideas that are appropriate for the discipline and genre
5. **Visuals/Graphics** Formal and informal rules inherent in the expectations for visual/graphical display including: proper layout, titling and labeling, and appropriate content	Uses visuals/graphics that skillfully communicate meaning to readers with clarity	Uses visuals/graphics that generally convey meaning	Uses visuals/graphics that minimally convey meaning	Does not use visuals/graphics; or does not use them appropriately

Grades: Rubric for project report

1. Technical Content (40 points): This concerns the correctness of theory, validity of reasoning used, apparent understanding and grasp of the subject, etc. Are all major factors considered and a reasonably accurate evaluation of these factors presented?

2. Organization and Presentation (15 points): The description of the design as an instrument of communication is a strong factor on judging. Organization of written design, clarity, and inclusion of pertinent information are major factors.

3. Originality (15 points): The design proposal should avoid standard textbook information, and should show the independence of thinking or a fresh approach to the project. Does the method and treatment of the problem show imagination? Does the method show an adaptation or creation of automated design tools?

4.Practical Application and Feasibility (30 points): The proposal should present conclusions or recommendations that are feasible and practical, and not merely lead the evaluators into further difficult or insolvable problems.

Grades: Weighting and scale

Course Grade Weighting:

Homework:	30%
Quiz #1:	10%
Quiz #2:	15%
Quiz #3:	10%
1st Presentation project:	7% *(includes peer evaluation)*
2nd Presentation project:	14% *(includes peer evaluation)*
Final report project:	14% *(includes peer evaluation)*
Total	100%

Homework:	individual
Quiz:	individual
Project:	a team-work

Grade Scale:

97 – 100%	A+	4.0
93–97%	A	4.0
90–93%	A-	3.7
87–90%	B+	3.3
83–87%	B	3.0
80–83%	B-	2.7
77–80%	C+	2.3
73–77%	C	2.0
70–73%	C-	1.7
67–70%	D+	1.3
63–67%	D	1.0
60–63%	D-	0.7
Below 60%	F	0.0

Grades: Re-evaluations & Emergencies

Grades: Homework and exam re-grade requests must be made within 1 week of the date that the graded document was made available to the student. There will be no curving of the grades, but the instructor may choose to lower certain grade cutoffs if it is justified.

In the event of a major campus emergency, course requirements, deadlines, and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances. Changes in this course will be announced in MS Teams.

Notes to ensure success in this course

The notes are **not a cookbook of equations**. If you understand the *physics* in the models you are using, you will succeed. We will provide some design recommendations, while you develop your critical thinking. If you are confused on by the diverse notations used in the literature, or confused by contradictory statements from the literature please see me.

In this course you are **expected to be engineer**. I value your understanding of the approach more than the validity of the solution. **Little credit** will be given if the solution demonstrates that you are simply plugging into equations you do not understand. There is **no substitute for** an **understanding** of the material presented in class. **If you are confused** on concepts in class **please see the TAs and me**.

There are many different texts available on propulsion. Although you are encouraged to read on subjects discussed in class **supplemental texts can't substitute for an understanding of the notes proposed and discussed in class**.

Grades: Attendance and general policy

Attendance: Although attendance is not required, students should attend all classes to receive the full benefit of the lectures.

Cheating will be prosecuted according to Purdue University policy

You may have other courses with a higher apparent workload in any given week.

Don’t fall behind in this class; it is difficult to catch up.

Remain passionate about propulsion

Grades: Academic integrity

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing **integrity@purdue.edu** or by calling 765-494-8778. While information may be submitted anonymously, the more information that is submitted provides the greatest opportunity for the university to investigate the concern

Purdue Honors Pledge: “As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue.”

Copyright

Among the materials that may be protected by copyright law are the lectures, notes, and other material presented in class or as part of the course. Always assume the materials presented by an instructor are protected by copyright unless the instructor has stated otherwise. Students enrolled in, and authorized visitors to, Purdue University courses are permitted to take notes, which they may use for individual/ group study or for other non-commercial purposes reasonably arising from enrollment in the course or the University generally.

Nondiscrimination

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. Link to Purdue's nondiscrimination policy statement.

Accessibility and Accommodations

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let us know so that we can discuss options. We also encourage you to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

Mental Health

If you find yourself beginning to feel some stress, anxiety and/or feeling a bit overwhelmed, tryWellTrack, https://purdue.welltrack.com/. Sign in and find information and tools at your fingertips, available to you at any time.

If you need support and information about options and resources, please see the Office of the Dean of Students, http://www.purdue.edu/odos, for drop-in hours (M-F, 8 am - 5 pm).

If you are struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available.

For help, such individuals should contact Counseling and Psychological Services (CAPS) at (765)494-6995 and http://www.purdue.edu/caps/ during and after hours, on weekends and holidays, or by going to the CAPS office of the second floor of the Purdue University Student Health Center (PUSH) during business hours.